



Assignment of master's thesis

Title: Automated Security Auditing of Ethereum Smart Contracts
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Branch / specialization: Knowledge Engineering
Department: Department of Applied Mathematics
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Instructions

Smart contract security audits are critical for preventing vulnerabilities that can lead to significant financial losses. Audits relying on manual code review are time-consuming and may miss still miss vulnerabilities. Recent advances in Large Language Models (LLMs) have demonstrated potential for automated reasoning and code analysis, opening new possibilities for AI-assisted security auditing.

The goal of this thesis is to investigate the application of LLMs for automated security auditing of Ethereum smart contracts and develop a prototype system that enhances the efficiency and coverage of vulnerability detection.

Instructions:

- Study Ethereum smart contract security, common vulnerability patterns, and existing auditing methodologies.
- Conduct a state of the art review on the integration of Large Language Models for automated reasoning and the development of AI-based auditing agents.
- Design a system that integrates an LLM into the security auditing workflow.
- Implement a prototype to perform automated vulnerability detection in smart contract audits.
- Evaluate and compare the implemented system with existing auditing tools and frameworks in terms of accuracy, coverage, and efficiency. Test the system on a dataset of real-world smart contracts with known vulnerabilities.
- Discuss the obtained results, limitations, and potential improvements of LLM-based auditing systems.